

Optimal Stopping Theory for Speculative Decoding Under Communication Constraints

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Abstract—Speculative decoding accelerates LLM inference by having a small draft model generate candidate tokens that a larger target model verifies. When draft and target reside on separate devices connected by an unreliable network, draft length selection becomes a stochastic optimization problem: longer speculation amortizes communication cost but suffers exponentially decaying acceptance, while shorter speculation maintains high acceptance at the expense of frequent transmissions. We formalize this tradeoff as a ratio-type optimal stopping problem. We prove that the optimal draft length k^* is a finite threshold, monotonically non-decreasing in delay, that scales only *logarithmically* with communication delay, and that a sharp *phase transition* exists at a critical delay d_c below which single-token speculation is always optimal. For Markov-modulated channels, the optimal policy is a state-dependent threshold. We propose UCB-SpecStop, an online algorithm achieving $O(\sqrt{K_{\max}T \log T})$ regret when system parameters are unknown. Hardware validation on an edge and cloud testbed (Jetson Orin Nano Super and RTX 3090) with Qwen2.5 and Llama 3 draft/target pairs confirms the predicted phase transitions, near-oracle strategy performance, and sublinear regret growth.

Index Terms—Speculative decoding, optimal stopping, communication uncertainty, large language models, distributed inference

I. INTRODUCTION

Large language models (LLMs) are increasingly deployed in latency-sensitive applications, from real-time dialogue systems to autonomous vehicle planning, where inference speed directly impacts user experience. Speculative decoding [1], [2] has emerged as a principled acceleration technique: a lightweight *draft model* generates k candidate tokens that a larger *target model* verifies in parallel, preserving the exact target distribution while potentially achieving up to $k\times$ speedup.

In centralized deployments (single machine or tightly coupled cluster), the draft length k is typically a fixed hyperparameter tuned offline [1], or adapted based on model confidence [3]. However, a growing class of deployments places the draft model on a resource-constrained *edge device* (e.g., mobile phone, IoT gateway) while the target model remains on a *cloud server* [4], [5]. In such distributed settings, each verification round incurs a *stochastic communication cost*, including network latency, jitter, and potential packet loss, which existing analyses entirely ignore.

This omission matters because it creates a fundamental, unresolved tradeoff. Speculating longer ($k \uparrow$) amortizes communication overhead across more tokens, but the acceptance probability decays exponentially (α^k), wasting

computation. Speculating shorter ($k \downarrow$) maintains high acceptance rates but multiplies communication rounds. While recent systems work [6]–[9] provides engineering evidence and profiling-based heuristics for communication-aware speculation, a closed-form structural characterization of the optimal draft length under stochastic communication, with provable regret guarantees for the online learning variant, has not been established. We close this gap with a tractable analytical model.

This paper addresses the following key questions:

- *Q1: What is the structure of the optimal speculation length under communication constraints?* Does an optimal threshold exist, and how does it depend on delay?
- *Q2: How should the strategy adapt to time-varying network conditions?* Can we characterize the value of observing the network state?
- *Q3: How can we learn the optimal strategy online when system parameters are unknown?*

We address these questions by formalizing the draft length decision as an *optimal stopping problem* [10]: at each draft step, the agent decides whether to continue generating tokens (diminishing marginal returns) or stop and transmit (stochastic communication cost). This formulation enables rigorous structural analysis using tools from sequential decision theory and lattice optimization. The contributions are summarized as follows:

- *Analytical formulation.* We formulate distributed speculative decoding draft length selection as a ratio-type optimal stopping problem under stochastic communication delay (§III). Under a tractable geometric-acceptance model, this formulation admits closed-form structural results that complement profiling-based and system-level treatments in recent edge-cloud speculative decoding work [6]–[8].
- *Structural theory with non-obvious insights.* We prove that the optimal policy is a threshold strategy (Theorem 2) and discover that the optimal draft length scales only *logarithmically* with communication delay (Theorem 5), a counter-intuitive result implying that even large increases in network latency warrant only modest increases in speculation length. We also identify a *phase transition* at a critical delay d_c (§IV).
- *Online learning with guarantees.* For unknown environments, we design UCB-SpecStop with a ratio-of-sums estimator aligned to (13), achieving $O(\sqrt{K_{\max}T \log T})$

regret (Theorem 7). The contextual variant maintains separate statistics per network state (§V).

- *Actionable design guidelines.* We derive closed-form expressions for the critical delay d_c and optimal draft length k^* , providing system designers with concrete formulas to configure distributed speculative decoding without exhaustive search.
- *Calibration-driven deployment.* For real systems, we outline a procedure that replaces idealized one-way delay and a single acceptance parameter with *measured* round-trip communication time and an *empirical* prefix acceptance curve, aligning hardware evaluation with the ratio-of-sums objective.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system model and optimal stopping formulation. Section IV develops the main theoretical results. Section V introduces the online learning algorithm. Section VI reports hardware-only experimental validation on an edge and cloud testbed. Section VII concludes the paper.

II. RELATED WORK

This section reviews four lines of related work: speculative decoding methods, distributed LLM inference systems, communication-aware and adaptive drafting systems, and optimal stopping theory. For each area, we highlight the gap that our work addresses.

A. Speculative Decoding

Speculative decoding has been extensively studied as a lossless acceleration technique for LLM inference. The draft-and-verify paradigm traces back to Stern et al. [11], who proposed blockwise parallel decoding that predicts multiple future positions and validates the longest acceptable prefix. Leviathan et al. [1] and Chen et al. [2] formalized speculative decoding with rejection sampling that provably preserves the target distribution. Subsequent work improves the draft mechanism along several dimensions: SpecInfer [12] introduces tree-structured speculation; EAGLE [3] leverages feature-level uncertainty for confidence-based adaptive drafting; Medusa [13] replaces the separate draft model with multiple decoding heads attached to the target model; and REST [14] uses retrieval to construct draft sequences. Online Speculative Decoding [15] adapts the draft model itself to improve acceptance rates over time.

Most centralized speculative decoding methods treat communication overhead as negligible; recent edge-cloud extensions (reviewed in §II-D) introduce network awareness primarily through profiling and engineering heuristics. Our work complements these systems by providing a closed-form structural analysis of the draft-length tradeoff and regret guarantees for online learning under a simplified stochastic delay model.

B. Distributed LLM Inference

Distributed LLM inference systems have emerged to improve hardware utilization by disaggregating inference phases across machines, but none provides theoretical guarantees for speculation strategy under communication uncertainty. Splitwise [4] disaggregates the prefill and decode phases across machines to improve hardware utilization. DistServe [5] further optimizes goodput by co-locating complementary workloads. These systems account for inter-machine communication at the engineering level.

While these systems address communication overhead implicitly through scheduling heuristics, they do not employ speculative decoding or provide theoretical guarantees for inference strategy selection under uncertain communication. Our work bridges speculative decoding theory with distributed inference by explicitly modeling communication randomness.

C. Optimal Stopping Theory

Optimal stopping theory provides a mature mathematical framework for sequential decisions under uncertainty, but has not been applied to the speculative decoding setting. Classical foundations [10], [16] include Wald’s sequential analysis [17] for hypothesis testing with sequential observations. Modern applications include optimal search, secretary problems, and financial option pricing. Topkis [18] established monotone comparative statics for optimization problems with supermodular structure.

The specific structure of our problem, namely a ratio objective with exponentially decaying denominator gains and linearly growing numerator costs, does not reduce to standard stopping problems and requires tailored analysis.

D. Communication-Aware and Adaptive Drafting

A growing body of work explicitly addresses speculative decoding in edge-cloud or communication-constrained settings. SLED [6] frames speculative decoding as an edge and server orchestration problem with dynamic drafting and timeouts under network uncertainty. Venkatesha et al. [9] build a real edge-cloud speculative decoding system with early exits and preemptive drafting, demonstrating cost-effective deployment. FlexSpec [7] introduces channel-aware adaptive speculation with frozen drafts and evolving targets. ConfigSpec [8] profiles configurations including speculative length K^* , draft model choice, quantization, and device placement for distributed serving.

On the algorithmic side, SpecDec++ [19] uses a learned acceptance predictor to adapt candidate lengths; its probability threshold can be viewed as an empirical analogue of our marginal-cost crossing condition (Corollary 1), specialized to content-dependent acceptance. TETRIS [20] studies batch speculative decoding and shows that effective draft depth interacts with batching and hardware saturation; our framework could incorporate this by letting verification cost depend on batch size $c_v(b)$, which we leave for future work. Batch speculative decoding with correctness guarantees [21] highlights

synchronization overheads from ragged acceptance in batched settings.

Our work differs from these systems in providing a closed-form structural theory (threshold optimality, monotonicity, phase transition, and logarithmic scaling) and regret guarantees for online learning under a simplified stochastic-delay model. We view our analytical results and these systems' empirical evidence as *complementary*: our closed-form d_c and $k^*(d)$ formulas give a fast configuration prior that systems such as ConfigSpec [8] can refine through profiling, while platforms such as SLED [6] provide the deployment substrate on which our policies can be evaluated.

III. SYSTEM MODEL AND PROBLEM FORMULATION

This section formalizes the distributed speculative decoding problem. Fig. 1 illustrates the system architecture and the fundamental tradeoff that motivates our formulation. The edge device hosts a lightweight draft model M_d that generates k candidate tokens, which are then transmitted over an unreliable network link to a cloud server hosting the target model M_t for parallel verification. The central challenge is selecting the draft length k : a short draft maintains high acceptance probability ($\alpha^k \approx 1$) but requires frequent communication rounds, while a long draft amortizes the stochastic communication cost $2D$ but suffers exponentially decaying acceptance ($\alpha^k \rightarrow 0$). As shown in the bottom of Fig. 1, this tradeoff produces a U-shaped cost-per-token curve $C(k, d)$ whose minimum defines the optimal draft length k^* .

We formalize this architecture (§III-A), introduce the token acceptance model (§III-B), define the cost-per-token objective (§III-C), and cast the problem as an optimal stopping problem (§III-D). The key result of this section is Definition 1, which establishes the formal framework for all subsequent theoretical analysis.

A. Distributed Inference Architecture

We consider a two-tier distributed inference system illustrated conceptually as follows.

(1) *Edge device*. Hosts the draft model M_d (e.g., a 7B-parameter model). Each draft token requires $c_d > 0$ time units to generate.

(2) *Cloud server*. Hosts the target model M_t (e.g., a 70B-parameter model). Verifying k draft tokens and generating one bonus token requires $(k + 1) \cdot c_v$ time units, where $c_v > 0$ is the per-token verification cost. Verification proceeds in parallel across all k tokens [1].

(3) *Communication channel*. The edge and cloud communicate over a channel with random one-way delay $D \geq 0$. We denote the round-trip cost as $2D$. The delay distribution may depend on an observable network state $s \in \mathcal{S}$.

B. Token Acceptance Model

We model the draft-target acceptance process following the analysis in [1]. Let $\alpha \in (0, 1)$ denote the probability that a single draft token is accepted by the target model's rejection sampling scheme.

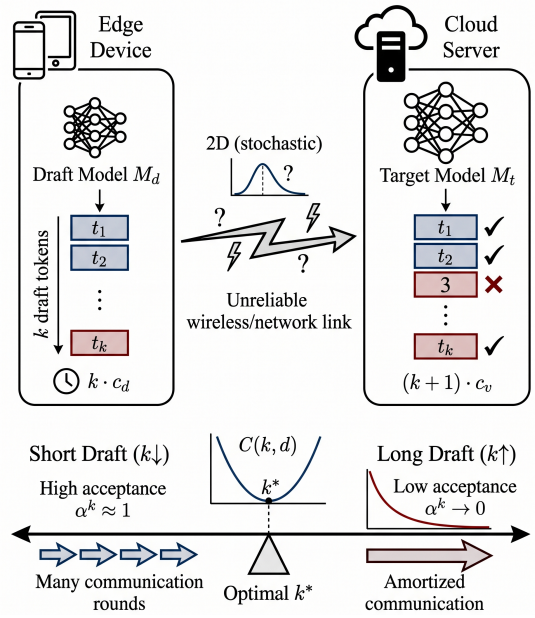


Fig. 1. Distributed speculative decoding architecture and the computation and communication tradeoff. *Top*: The edge device generates k draft tokens (cost $k \cdot c_d$), transmits them over an unreliable channel (round-trip cost $2D$), and the cloud server verifies all tokens in parallel (cost $(k + 1) \cdot c_v$). *Bottom*: Short drafts yield high acceptance ($\alpha^k \approx 1$) but incur many communication rounds; long drafts amortize communication but suffer exponentially decaying acceptance ($\alpha^k \rightarrow 0$). The optimal draft length k^* minimizes the U-shaped cost-per-token $C(k, d)$.

Assumption 1 (Geometric Acceptance). *Token acceptance events are conditionally independent across positions. The probability that all k consecutive draft tokens are accepted is α^k .*

Assumption 1 is standard in the speculative decoding literature [1], [2] and captures the essential exponential decay structure. While real acceptance rates exhibit positional dependence [3], this model enables tractable closed-form analysis. We discuss extensions to content-dependent acceptance in Section VII (Future Work).

Under Assumption 1, the expected number of accepted tokens (including the bonus token generated by the target model upon the first rejection) from a draft of length k is:

$$B(k) \triangleq \mathbb{E}[A(k)] = \frac{1 - \alpha^{k+1}}{1 - \alpha} \quad (1)$$

We introduce the notation $B(k)$ for conciseness throughout the paper.

C. Cost-Per-Token Objective

For a speculation round with draft length k , the total elapsed time is:

$$T(k, D) = \underbrace{k \cdot c_d}_{\text{draft generation}} + \underbrace{2D}_{\text{round-trip comm.}} + \underbrace{(k + 1) \cdot c_v}_{\text{parallel verification}} \quad (2)$$

We measure performance by the expected time per accepted token:

$$C(k, d) = \frac{N(k, d)}{B(k)} = \frac{k(c_d + c_v) + 2d + c_v}{(1 - \alpha^{k+1})/(1 - \alpha)} \quad (3)$$

where $N(k, d) \triangleq k(c_d + c_v) + 2d + c_v$ denotes the total cycle cost and $d = \mathbb{E}[D]$ in the deterministic approximation. The goal is to find the draft length k minimizing $C(k, d)$.

D. Optimal Stopping Formulation

We now cast draft length selection as a sequential decision problem.

Definition 1 (Optimal Stopping for Speculative Decoding). *At each discrete step $n = 1, 2, 3, \dots$, the agent has generated n draft tokens. The agent observes the network state $s_n \in \mathcal{S}$ and selects an action:*

- *Continue ($a_n = 0$): Generate token $n+1$, incurring cost c_d .*
- *Stop ($a_n = 1$): Transmit n tokens for verification, incurring communication cost $2D(s_n)$ and verification cost $(n+1)c_v$.*

The objective is to find a stopping rule τ minimizing:

$$\min_{\tau} \frac{\mathbb{E}[\tau \cdot c_d + 2D(s_{\tau}) + (\tau + 1) \cdot c_v]}{\mathbb{E}[B(\tau)]} \quad (4)$$

This is a *ratio-type* optimal stopping problem [10]. The standard approach converts it to an additive problem via Lagrange multipliers: for a given target rate λ , find τ minimizing $\mathbb{E}[N(\tau, D) - \lambda \cdot B(\tau)]$, then search for λ^* such that the constraint binds.

In summary, we have established a precise mathematical formulation (Definition 1) that captures the computation and communication tradeoff in distributed speculative decoding. The next section derives structural properties of the optimal solution.

IV. THEORETICAL RESULTS

This section presents our main theoretical contributions. We proceed in three stages of increasing generality: deterministic delay (§IV-A), stochastic delay (§IV-B), and Markov-modulated channels (§IV-C). We then present the paper’s most surprising result, a phase transition and logarithmic scaling law (§IV-D), followed by a characterization of the value of network state information (§IV-E). The key takeaways are:

- The optimal draft length k^* always exists and is finite (Theorem 1).
- k^* is monotonically non-decreasing in delay (Theorem 2).
- k^* grows only logarithmically with delay (Theorem 5).

A. Deterministic Communication Delay

We begin with the simplest setting: the communication delay is a known constant $d \geq 0$. This baseline analysis reveals the fundamental structure that persists in more general settings.

Theorem 1 (Existence and Finiteness). *Under Assumption 1 with $\alpha \in (0, 1)$, $c_d > 0$, $c_v \geq 0$, and $d \geq 0$, the cost function $C(k, d)$ achieves its minimum at a finite $k^* \in \mathbb{N}^+$.*

Proof Sketch. We establish the result by analyzing the boundary behavior of $C(k, d)$. The numerator $N(k, d) = k(c_d + c_v) + 2d + c_v$ grows as $\Theta(k)$, while the denominator $B(k) \rightarrow (1 - \alpha)^{-1}$ is bounded. Hence $C(k, d) \rightarrow \infty$ as $k \rightarrow \infty$. Then, we have $C(1, d) = (c_d + 2d + 2c_v)/(1 + \alpha) < \infty$. Since $C : \mathbb{N}^+ \rightarrow \mathbb{R}_{>0}$ is bounded below by zero and diverges, it attains its infimum at some finite k^* . $\square$

The finiteness of k^* arises from the tension between two forces: the communication cost $2d$ favors large k (amortization), while the exponential acceptance decay α^k penalizes large k (diminishing returns). Neither force dominates asymptotically, and their balance determines k^* .

We now characterize how k^* responds to changes in communication delay. The following result answers **Q1** from the Introduction.

Theorem 2 (Monotonicity). *The optimal draft length $k^*(d)$ is monotonically non-decreasing in d : for $d_1 \leq d_2$, we have $k^*(d_1) \leq k^*(d_2)$.*

Proof Sketch. We apply Topkis’ monotone comparative statics theorem [18]. Define $f(k, d) = -C(k, d)$. We verify the increasing differences condition: for $k_2 > k_1$,

$$\frac{\partial}{\partial d} [C(k_1, d) - C(k_2, d)] = \frac{2(B(k_2) - B(k_1))}{B(k_1)B(k_2)} > 0$$

The positivity follows because $B(\cdot)$ is strictly increasing. This establishes that the *relative* advantage of larger k grows with d . By Topkis’ theorem, the maximizer of $f(k, d)$, equivalently the minimizer of $C(k, d)$, is non-decreasing in d . $\square$

When communication is expensive (large d), the fixed cost $2d$ dominates the per-cycle budget. The agent should “batch” more draft tokens per round to amortize this fixed cost, accepting the reduced marginal acceptance probability. This is analogous to the economic principle that higher fixed costs favor larger production batches.

The following corollary provides an operational stopping rule.

Corollary 1 (Marginal Cost Stopping Rule). *The optimal k^* is the smallest integer satisfying:*

$$C(k, d) \leq \frac{c_d + c_v}{\alpha^{k+1}} \quad (5)$$

Equivalently: stop when the current average cost per token falls below the marginal cost of generating one additional draft token.

Proof. The condition $C(k+1, d) \geq C(k, d)$ simplifies to $(c_d + c_v) \cdot B(k) \geq N(k, d) \cdot \alpha^{k+1}$, which rearranges to (5). The derivation uses $N(k+1, d) = N(k, d) + (c_d + c_v)$ and $B(k+1) = B(k) + \alpha^{k+1}$. $\square$

Condition (5) is the discrete analog of the classical “marginal cost equals average cost” optimality condition. The

marginal cost $\frac{c_d+c_v}{\alpha^{k+1}}$ grows exponentially (reflecting worsening acceptance odds), while the average cost $C(k, d)$ first decreases (amortizing the communication fixed cost) then increases. Their crossing defines k^* . This answers **Q1**: the optimal strategy has an intuitive threshold structure with a precise mathematical characterization.

B. Stochastic Communication Delay

We now relax the deterministic delay assumption. Let D be a non-negative random variable with mean μ_D and the stopping decision is committed *before* observing the delay realization (the “commit-before-observing” model).

Theorem 3 (Mean-Sufficiency under Commit-Before-Observing). *Under the commit-before-observing model, the optimal draft length depends on the delay distribution only through its mean:*

$$k^* = \arg \min_{k \geq 1} \frac{k(c_d + c_v) + 2\mu_D + c_v}{B(k)} \quad (6)$$

Proof Sketch. Taking expectations over D with fixed k yields $\mathbb{E}[C(k, D)] = C(k, \mu_D)$, reducing to the deterministic case. $\square$

Theorem 3 shows that under commitment, only the mean delay affects the optimal strategy, while higher moments are irrelevant. This *changes fundamentally* when the agent can observe network state before deciding, as we show next. The gap between these two information structures quantifies the value of real-time network monitoring.

C. State-Dependent Optimal Policy

We now model time-varying network conditions. The network state $\{s_t\}$ evolves as a Markov chain, and the agent observes s_n after generating draft token n , using this information to decide whether to stop. This addresses **Q2**.

Assumption 2 (Markov-Modulated Channel). *The network state $\{s_t\}_{t \geq 1}$ is a finite-state Markov chain on $\mathcal{S} = \{1, \dots, S\}$ with transition matrix P . We assume:*

- (a) Monotone mean delay: $d(s) = \mathbb{E}[D \mid s_t = s]$ is non-decreasing in s (states ordered from low to high delay).
- (b) Stochastic monotonicity of P : for every non-decreasing function $h : \mathcal{S} \rightarrow \mathbb{R}$, the map $s \mapsto \sum_{s'} P(s' \mid s) h(s')$ is non-decreasing in s . Equivalently, $P(\cdot \mid s)$ is stochastically increasing in s in the usual stochastic order.

Remark 1. Condition (b) is standard for monotone MDPs [18] and holds, for example, for birth and death chains modeling congestion levels and for any tridiagonal P whose row distributions are stochastically ordered. Without (b), worse states can transition to better states faster than better states do, which can break the monotonicity of value functions.

Theorem 4 (State-Dependent Threshold Policy). *Under Assumptions 1 and 2, the optimal policy is a state-dependent threshold: there exists $k^* : \mathcal{S} \rightarrow \mathbb{N}^+$ such that*

- 1) in state s , the optimal action is to stop at draft length $k^*(s)$;

- 2) $k^*(s)$ is non-decreasing in s .

Proof. We apply the Dinkelbach transformation [22] to convert the ratio objective in (4) to an equivalent additive stopping problem.

Step 1 (parametric reformulation). For $\lambda \geq 0$, define the λ -penalized stopping payoff in (7). Let $J(\lambda) = \min_{\tau} \mathbb{E}[g(\tau, s_{\tau}; \lambda)]$. By standard fractional programming [22], $J(\cdot)$ is continuous and strictly decreasing on $[0, \infty)$, and the unique root λ^* of $J(\lambda) = 0$ equals the optimal ratio $\min_{\tau} \mathbb{E}[N(\tau)]/\mathbb{E}[B(\tau)]$. Any optimal stopping rule for the additive problem at λ^* is optimal for the ratio problem.

Step 2 (Bellman equation and finite threshold). For fixed λ , the Bellman equation (8) admits a unique bounded solution because the per-stage cost is bounded and stopping is always feasible. As $n \rightarrow \infty$, the stopping payoff $g(n, s; \lambda)$ grows linearly in n since $B(n) \rightarrow (1 - \alpha)^{-1}$ is bounded, while continuing yields at most a vanishing α^{n+1} marginal reward. Hence for each s , stopping is optimal for all sufficiently large n , so a finite threshold $k^*(s; \lambda) < \infty$ exists.

For $\lambda \geq 0$, define the λ -penalized stopping payoff after n draft tokens in state s :

$$g(n, s; \lambda) \triangleq N(n, d(s)) - \lambda B(n). \quad (7)$$

Step 3 (monotonicity in s via stochastic dominance). For fixed λ , define the continuation value

$$u_{\lambda}(n, s) \triangleq c_d - \lambda \alpha^{n+1} + \sum_{s'} P(s' \mid s) V_{\lambda}(n+1, s').$$

The Bellman equation is

$$V_{\lambda}(n, s) = \min\{g(n, s; \lambda), u_{\lambda}(n, s)\}. \quad (8)$$

Here g and u_{λ} share the same (time) units: continuing adds draft cost c_d and increases expected accepted tokens by α^{n+1} , valued at λ per token.

We show by backward induction on n that $V_{\lambda}(n, s)$ is non-decreasing in s . Truncate at horizon $N_{\max}$ and let $N_{\max} \rightarrow \infty$; the truncation limit is justified by the boundedness of V_{λ} . *Base case:* at $n = N_{\max}$, the agent must stop, so $V_{\lambda}(N_{\max}, s) = g(N_{\max}, s; \lambda) = N_{\max}(c_d + c_v) + 2d(s) + c_v - \lambda B(N_{\max})$, which is non-decreasing in s by Assumption 2(a). *Inductive step:* suppose $V_{\lambda}(n+1, \cdot)$ is non-decreasing in s . Then by Assumption 2(b), $\sum_{s'} P(s' \mid s) V_{\lambda}(n+1, s')$ is non-decreasing in s , hence $u_{\lambda}(n, s)$ is non-decreasing in s . Since $g(n, s; \lambda)$ is also non-decreasing in s by (a), the minimum (8) preserves monotonicity.

Step 4 (threshold monotonicity in s). Define the stopping advantage $\Delta_{\lambda}(n, s) \triangleq u_{\lambda}(n, s) - g(n, s; \lambda)$. Continuing is optimal iff $\Delta_{\lambda}(n, s) \leq 0$. The discrete derivative of g in n is $g(n+1, s; \lambda) - g(n, s; \lambda) = (c_d + c_v) - \lambda \alpha^{n+1}$, which is independent of s and non-decreasing in n . By Step 3 and Assumption 2(b), the continuation value's discrete derivative inherits monotonicity in s . A standard monotone comparative statics argument (Topkis [18]) on $\Delta_{\lambda}(n, s)$, which has decreasing differences in (n, s) under (a) and (b), implies that the smallest n at which $\Delta_{\lambda}(n, s) \leq 0$, i.e., $k^*(s; \lambda)$, is non-decreasing in s .

Step 5 (transfer to the ratio problem). Setting $\lambda = \lambda^*$ from Step 1, the policy $k^*(s) \triangleq k^*(s; \lambda^*)$ is optimal for the original ratio problem (4) and remains non-decreasing in s . $\square$

In (8), the continuation term $c_d - \lambda \alpha^{n+1}$ admits a marginal interpretation: one more draft token costs c_d but yields an expected α^{n+1} additional accepted tokens, each worth λ under the Dinkelbach parameter. When $c_d > \lambda \alpha^{n+1}$, continuing raises cost faster than it “earns” in λ -units, favoring stopping.

Theorem 4 shows that a distributed inference system must monitor network quality indicators such as RTT estimates and congestion signals. It should use these metrics to dynamically adjust speculation length, and speculate more aggressively when high-delay periods are detected.

D. Phase Transition and Logarithmic Scaling

The preceding results establish that k^* increases with d . A natural follow-up question is: *how fast?* The following theorem is our most surprising result and shows the growth is remarkably slow.

Theorem 5 (Phase Transition and Scaling Law). *Define the critical delay:*

$$d_c = \frac{(c_d + c_v)(1 + \alpha)}{2\alpha^2} - \frac{c_d + 2c_v}{2} \quad (9)$$

Then:

- 1) *Sub-critical regime* ($d < d_c$): $k^* = 1$.
- 2) *Phase transition* ($d = d_c$): k^* jumps from 1 to 2.
- 3) *Asymptotic regime* (large d): $k^* = \Theta\left(\frac{\log d}{\log(1/\alpha)}\right)$.

Proof Sketch. From (5), $k^*=1$ iff $C(1, d) \leq (c_d + c_v)/\alpha^2$. Expanding:

$$\frac{c_d + 2d + 2c_v}{1 + \alpha} \leq \frac{c_d + c_v}{\alpha^2}$$

Solving for d yields (9).

For $d \gg c_d, c_v$, the optimality condition $C(k, d) \approx (c_d + c_v)/\alpha^{k+1}$ gives $2d \cdot \alpha^{k+1} \approx (c_d + c_v)/(1 - \alpha)$. Taking logarithms:

$$k^* \approx \frac{\log(2d(1 - \alpha)/(c_d + c_v))}{\log(1/\alpha)} - 1$$

$\square$

Theorem 5 is the central insight of this paper. It reveals three non-obvious facts:

(1) Doubling communication delay adds only $\approx 1/\log(1/\alpha)$ tokens to the optimal draft length. For typical $\alpha = 0.7$, this is merely ≈ 2.8 extra tokens. The intuition is that exponential acceptance decay provides a natural “damping” that prevents the optimal response from overreacting to delay increases.

(2) Below d_c , communication is cheap enough that single-token speculation (maximum verification frequency) is optimal regardless of further delay reduction. This provides a concrete threshold for system designers: if $d < d_c$, there is no benefit from multi-token speculation.

(3) The logarithmic scaling means that even in extremely high-delay scenarios (e.g., satellite links), the optimal draft

length remains moderate (e.g., $k^* \approx 10$ to 15 for $d = 500$ ms, $\alpha = 0.7$). System implementations need not support arbitrarily long speculation buffers.

E. Value of Network State Information

Theorem 4 shows that observing network state improves the policy. We now quantify this improvement, answering the second part of **Q2**.

Theorem 6 (Value of Information). *Let k_{blind}^* be the optimal fixed strategy and $k^*(\cdot)$ the state-dependent policy. The value of information is:*

$$\text{VOI} = C(k_{\text{blind}}^*, \mu_D) - \mathbb{E}_s[C(k^*(s), d(s))] \geq 0 \quad (10)$$

Moreover:

- 1) $\text{VOI} = 0$ when $\text{Var}(D) = 0$.
- 2) $\text{VOI} > 0$ when the delay variance is large enough for different states to fall in different threshold regions (i.e., cross a phase transition boundary from Theorem 5).

Proof Sketch. Since $C(k, d)$ is linear in d for fixed k , we have $\mathbb{E}_s[C(k_{\text{blind}}^*, d(s))] = C(k_{\text{blind}}^*, \mu_D)$. The state-dependent policy satisfies $C(k^*(s), d(s)) \leq C(k_{\text{blind}}^*, d(s))$ pointwise, with strict inequality when $k^*(s) \neq k_{\text{blind}}^*$. $\square$

Our structural results rely on the acceptance model to varying degrees. Theorem 1 (existence/finiteness) requires only that $B(k)$ remain bounded as $k \rightarrow \infty$, i.e., that acceptance probabilities decay fast enough. Theorem 2 requires $B(k)$ strictly increasing in k , hence positive per-position acceptance. Theorem 5 and the logarithmic envelope use exponential decay α^k ; for heavier-tailed acceptance (e.g., $q_k \propto (1+k)^{-\beta}$), scaling becomes super-logarithmic. Corollary 1 holds whenever $C(k, d)$ is eventually increasing in k (quasi-convexity), which is typical when marginal acceptance decays faster than marginal draft cost grows. Extensions to content-dependent, non-stationary acceptance are important future work.

This theorem provides a decision criterion for whether to invest in network monitoring infrastructure: if the network delay variance is small (e.g., wired datacenter interconnect), a fixed k suffices; if the variance is large (e.g., cellular networks), state-adaptive strategies provide measurable benefit.

In summary, this section establishes that the optimal speculation strategy under communication constraints is a threshold policy with logarithmic delay sensitivity and a sharp phase transition. The next section addresses the practical challenge of unknown system parameters.

V. ONLINE LEARNING ALGORITHM

This section addresses **Q3**: learning the optimal draft length when both the acceptance rate α and delay distribution are unknown, a practical scenario arising when deploying on new hardware or connecting to unfamiliar network environments. The key technical challenge is that our objective (13) is a *ratio of expectations* rather than a simple expected reward, which invalidates standard bandit algorithms designed for additive objectives. Naively minimizing the per-round ratio N_t/A_t

optimizes a different criterion (due to Jensen's inequality), necessitating a ratio-of-sums estimator with carefully designed confidence bounds.

We formulate the problem as a multi-armed bandit (§V-A), present the UCB-SpecStop algorithm (§V-B), prove its regret guarantee (§V-C), and extend it to the contextual setting (§V-D). The main result is Theorem 7, which establishes $O(\sqrt{K_{\max} T \log T})$ regret despite the non-standard ratio structure.

A. Bandit Formulation

The theoretical results of Section IV assume known α , c_d , c_v , and delay distribution. In practice, these parameters are difficult to estimate *a priori*: the acceptance rate α depends on the specific draft and target model pair and input distribution, while network delay statistics vary across deployment environments and time. We therefore seek an online algorithm that learns the optimal draft length through repeated interaction.

At each decoding round $t = 1, \dots, T$:

- 1) The agent selects $k_t \in \mathcal{K} = \{1, \dots, K_{\max}\}$.
- 2) Nature reveals the realized numerator and denominator

$$N_t = k_t(c_d + c_v) + 2D_t + c_v, \quad (11)$$

$$A_t = \text{number of accepted tokens in the round}, \quad (12)$$

with D_t the realized delay. The per-round *ratio* N_t/A_t is a random variable whose expectation generally differs from $\mathbb{E}[N_t]/\mathbb{E}[A_t]$ (Jensen's inequality).

A subtle but crucial point is the choice of optimization criterion. The design objective is the *ratio of expectations*:

$$C(k) \triangleq \frac{\mathbb{E}[N_t \mid k_t = k]}{\mathbb{E}[A_t \mid k_t = k]}, \quad (13)$$

which aligns with (3) when (D_t, A_t) are drawn from the same generative model for each fixed k . Minimizing $\mathbb{E}[N_t/A_t \mid k_t = k]$ instead would optimize the *expectation of the ratio*, which is a different criterion that overweights rounds with few accepted tokens. We therefore use a *ratio-of-sums* estimator that directly targets (13).

Definition 2 (Regret). *Let $k^* \in \arg \min_{k \in \mathcal{K}} C(k)$. The cumulative regret is*

$$R(T) = \sum_{t=1}^T (C(k_t) - C(k^*)). \quad (14)$$

The regret definition charges the *expected* suboptimality gap $C(k_t) - C(k^*)$ at each round, even though the agent observes only stochastic realizations (N_t, A_t) . This is standard in the bandit literature [23] and measures the cumulative cost of exploration.

B. UCB-SpecStop Algorithm

The core design challenge is constructing a reliable confidence interval for the ratio estimator $\hat{C}(k) = S_N(k)/S_A(k)$. Unlike standard UCB1 where the sample mean directly estimates the objective, here the ratio of two random quantities requires joint concentration control.

Algorithm 1 UCB-SpecStop (ratio-of-sums)

Require: Arms $\mathcal{K} = \{1, \dots, K_{\max}\}$, confidence parameter $\beta > 0$, horizon T

Ensure: Estimated optimal draft length $\hat{k}^*$

- 1: **Initialize:** $S_N(k) \leftarrow 0$, $S_A(k) \leftarrow 0$, $T_k \leftarrow 0$ for all $k \in \mathcal{K}$
 - 2: **for** $t = 1, 2, \dots, T$ **do**
 - 3: **if** there exists k with $T_k = 0$ **then**
 - 4: $k_t \leftarrow$ any such k
 - 5: **else**
 - 6: $k_t \leftarrow \arg \min_{k \in \mathcal{K}} \left[\frac{S_N(k)}{S_A(k)} - \beta \frac{\sqrt{T_k \log t}}{S_A(k)} \right]$
 - 7: **end if**
 - 8: Play k_t ; observe N_t and A_t
 - 9: $S_N(k_t) \leftarrow S_N(k_t) + N_t$; $S_A(k_t) \leftarrow S_A(k_t) + A_t$;
 $T_{k_t} \leftarrow T_{k_t} + 1$
 - 10: **end for**
 - 11: **return** $\hat{k}^* \leftarrow \arg \min_{k \in \mathcal{K}} S_N(k)/S_A(k)$
-

For each arm k , we maintain cumulative statistics $S_N(k) = \sum_{t:k_t=k} N_t$ and $S_A(k) = \sum_{t:k_t=k} A_t$, and let T_k count how many times arm k is played. The ratio-of-sums estimator $\hat{C}(k) = S_N(k)/S_A(k)$ is consistent for $C(k)$ by the strong law of large numbers applied to numerator and denominator separately. Crucially, this estimator avoids the bias inherent in averaging per-round ratios $\frac{1}{T_k} \sum N_t/A_t$.

Algorithm 1 combines this estimator with a confidence width proportional to $\sqrt{T_k \log t}/S_A(k)$. The denominator $S_A(k)$ in the confidence width reflects that arms producing more accepted tokens per round provide more “information per play,” leading to faster concentration of the ratio estimate.

The selection rule in Line 6 can be understood as an optimistic estimate: the agent subtracts a confidence bonus from the cost estimate, selecting the arm with the lowest plausible cost. The $\sqrt{T_k \log t}$ term balances exploration (trying under-sampled arms) against exploitation (playing the empirically best arm). The division by $S_A(k)$ rather than T_k accounts for the heterogeneous information content across arms, because arms with higher acceptance rates (larger $B(k)$) accumulate S_A faster, tightening their confidence intervals more quickly.

C. Regret Guarantee

Assumption 3 (Bounded costs). *The delays are bounded: $D \leq D_{\max}$ almost surely. With $K_{\max} < \infty$, we have $N_t \leq N_{\max} \triangleq K_{\max}(c_d + c_v) + 2D_{\max} + c_v$. Moreover $A_t \geq 1$ almost surely (at least the bonus token is accepted each round), so $\mathbb{E}[A_t \mid k_t = k] \geq B_{\min} \triangleq 1$.*

The boundedness of $A_t \geq 1$ is guaranteed by the speculative decoding protocol: even if all k draft tokens are rejected, the target model generates one bonus token [1]. This structural property is essential because it ensures the ratio estimator's denominator never degenerates, avoiding the pathological behavior that plagues ratio bandits with potentially zero denominators.

Theorem 7 (Regret bound). *Under Assumption 3, the corrected UCB-SpecStop (Algorithm 1) satisfies, for suboptimality gaps $\Delta_k = C(k) - C(k^*)$,*

$$R(T) = O\left(K_{\max} \left(\frac{N_{\max}}{B_{\min}}\right)^2 \log T\right) \quad (15)$$

in the gap-dependent regime, and $R(T) = O(\sqrt{K_{\max} T \log T} \cdot N_{\max}/B_{\min})$ gap-free, where $B_{\min} = 1$.

Proof Sketch. Concentration for $\hat{C}(k) = S_N/S_A$ follows from Hoeffding bounds on S_N/T_k and S_A/T_k and the delta method: if $|S_N/T_k - \mathbb{E}[N | k]| \leq \varepsilon_N$ and $|S_A/T_k - \mathbb{E}[A | k]| \leq \varepsilon_A$, then

$$\left| \frac{S_N}{S_A} - \frac{\mathbb{E}[N | k]}{\mathbb{E}[A | k]} \right| \leq \frac{\varepsilon_N}{\mathbb{E}[A | k]} + \frac{\mathbb{E}[N | k] \varepsilon_A}{\mathbb{E}[A | k]^2} + O(\varepsilon^2).$$

Since $\mathbb{E}[A | k] \geq B_{\min}$, the ratio concentrates with constants inflated by $N_{\max}/B_{\min}$. The remainder follows the UCB1 template [23] with this modified width. $\square$

The $O(\sqrt{K_{\max} T \log T})$ gap-free bound carries a leading constant proportional to $N_{\max}/B_{\min}$. This ratio captures the “signal-to-noise” difficulty of the problem: large $N_{\max}$ (high-delay environments) increases cost variance, while small $B_{\min}$ (low acceptance) reduces the information gained per round. For typical ranges ($K_{\max} \leq 20$, $D_{\max}$ on the order of tens to hundreds of ms), $N_{\max}$ remains moderate.

The dependence on $K_{\max}$ reflects the cost of exploring all arms. However, Theorem 5 implies that only $O(\log d / \log(1/\alpha))$ arms are competitive in high-delay regimes, and the remaining arms have large gaps Δ_k and are quickly eliminated. This structural insight opens the door to tighter instance-dependent analyses that exploit the unimodal structure of $C(k, d)$, potentially reducing the effective arm count from $K_{\max}$ to $O(\log d)$.

D. Contextual Extension

When the network state s_t is observable (e.g., through RTT measurements or congestion indicators), the agent can leverage this side information to accelerate learning. The key idea is simple: maintain independent statistics per state, effectively running $|S|$ parallel bandit instances. This is justified by Theorem 4, which guarantees that the optimal arm $k^*(s)$ depends on the state, and sharing statistics across states would conflate different optimal policies.

We maintain independent (S_N, S_A, T) statistics per (k, s) and run the same lower-index rule within the active state:

Corollary 2. *Contextual UCB-SpecStop achieves $R(T) = O(|S| K_{\max} (N_{\max}/B_{\min})^2 \log T)$ gap-dependent and $R(T) = O(\sqrt{|S| K_{\max} T \log T} N_{\max}/B_{\min})$ gap-free, relative to the optimal state-dependent policy $k^*(\cdot)$.*

The multiplicative factor $|S|$ reflects the cost of learning a separate optimal arm in each state. In practice, $|S|$ is small (e.g., $|S| = 2$ for a good/bad channel model), so the overhead is modest. Furthermore, the monotonicity $k^*(s) \leq k^*(s')$ for $s \leq s'$ (Theorem 4) could be exploited to share information

Algorithm 2 Contextual UCB-SpecStop (ratio-of-sums)

Require: State space $\mathcal{S}$, arms $\mathcal{K}$, $\beta > 0$, horizon T

Ensure: State-dependent optimal policy $\hat{k}^*(s)$ for all $s \in \mathcal{S}$

```

1: Initialize:  $S_N(k, s) \leftarrow 0$ ,  $S_A(k, s) \leftarrow 0$ ,  $T_{k,s} \leftarrow 0$  for all  $(k, s)$ 
2: for  $t = 1, 2, \dots, T$  do
3:   Observe  $s_t$ 
4:   if some  $k$  has  $T_{k,s_t} = 0$  then
5:      $k_t \leftarrow$  any such  $k$ 
6:   else
7:      $k_t \leftarrow \arg \min_k [S_N(k, s_t)/S_A(k, s_t) - \beta \sqrt{T_{k,s_t} \log t} / S_A(k, s_t)]$ 
8:   end if
9:   Play  $k_t$ ; observe  $N_t, A_t$ ; update  $(S_N, S_A, T)_{k_t, s_t}$ 
10: end for
11: return  $\hat{k}^*(s) \leftarrow \arg \min_k S_N(k, s)/S_A(k, s)$ ,  $\forall s \in \mathcal{S}$ 
```

across adjacent states, though we leave this refinement for future work.

Theorem 6 provides a principled criterion: if the delay variance is small (all states yield the same k^*), the non-contextual Algorithm 1 suffices and converges faster (no state splitting). If the variance is large enough for states to straddle the phase transition d_c , contextual Algorithm 2 captures the VOI and achieves lower asymptotic cost. In practice, one can start with the non-contextual version and switch to contextual when observed delay variance exceeds a threshold derived from (9).

In summary, UCB-SpecStop addresses the practical challenge of unknown parameters by combining a ratio-of-sums estimator (aligned with the true objective (13)) with UCB-style exploration. The $O(\sqrt{K_{\max} T \log T})$ regret guarantee ensures that the cumulative cost of learning vanishes relative to the horizon T , while the contextual extension leverages network state information to match the performance of the oracle state-dependent policy from Theorem 4. This completes our answer to **Q3**. The next section validates both the theoretical predictions and the online algorithm through hardware measurements.

VI. EXPERIMENTAL VALIDATION

We validate the main theoretical predictions and UCB-SpecStop on a real edge and cloud testbed. To keep the main text focused, this section reports the three core rounds that directly map to the paper’s main contributions: R3 (phase transition and cost-vs.- k), R4 (strategy comparison), and R5 (online regret). Rounds R1 to R2 and R6 are deferred to the Appendix, together with R5 supplementary convergence/ β -sensitivity analyses. The appendix and code are available at the github repository ¹.

A. Experimental Setup

The edge runs on an NVIDIA Jetson Orin Nano Super executing the draft model; the cloud runs on an

¹<https://github.com/szpsunkk/UCB-SpecStop>

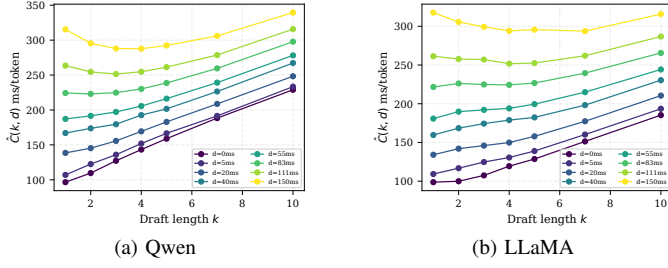


Fig. 2. Per-token cost $\hat{C}(k, d)$ vs. k for $d \in \{0, 5, 20, 40, 55, 83, 111, 150\}$ ms; minima highlighted.

NVIDIA RTX 3090 executing the target model. The two nodes communicate over the LAN with controllable one-way delay injected at the cloud interface via Linux `tc netem`. Bare-metal baseline RTTs measured during calibration (Round R1) are 10.01 ms (Qwen suite) and 9.02 ms (LLaMA suite), consistent with a lightly loaded gigabit link. We evaluate two draft and target pairs: (i) *Qwen suite*, draft Qwen/Qwen2.5-0.5B, target Qwen/Qwen2.5-7B-Instruct; (ii) *LLaMA suite*, draft Llama-3.2-1B-Instruct, target meta-llama/Llama-3.1-8B-Instruct. Per-token costs c_d, c_v and acceptance profiles are calibrated by direct measurement.

We report the *ratio-of-sums* per-token latency $\hat{C} = \sum_r T_r / \sum_r A_r$ (ms/token), aligned with the arm objective (13) and the ratio-of-sums analysis in Section V. Cumulative regret $R(t)$ in R3 uses the empirical oracle $C^*(d)$, the minimum ratio-of-sums cost across all fixed- k arms at the same delay, as the reference. Appendix B profiles per-position acceptance and shows a heavy head, $\hat{q}(1) = 0.462/0.382$ (Qwen/LLaMA), with a near-geometric tail $\alpha_{\text{geo}} \approx 0.83/0.85$ for $k \geq 2$, supporting the use of Assumption 1 for structural analysis.

B. Phase Transition and Cost Curves

Fig. 2 plots the measured per-token cost $\hat{C}(k, d)$ as a function of the draft length k over the full one-way delay grid, and Fig. 3 reports the empirical optimum $\hat{k}^*(d)$ together with the three theoretical predictions (geometric, calibrated-geometric, and empirical-prefix oracles).

Three observations match Theorem 5 (phase transition and logarithmic scaling) and the monotonicity result of Theorem 2. (i) The cost curves $\hat{C}(\cdot, d)$ are U-shaped and strictly convex around their empirical minimum for every d in the grid, so a unique $\hat{k}^*(d)$ is well defined. (ii) For both suites the sub-critical region $\hat{k}^*(d) = 1$ covers all small and moderate delays, and the first jump to $\hat{k}^* \geq 2$ occurs at a suite-specific critical delay: $d_c^{(\text{Qwen})} \approx 83$ ms and $d_c^{(\text{LLaMA})} \approx 111$ ms. (iii) Beyond the transition, $\hat{k}^*$ grows slowly: Qwen moves $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ between 55 and 150 ms; LLaMA moves $1 \rightarrow 4 \rightarrow 7$. Both trajectories are consistent with the $\lceil \log d / \log(1/\alpha) \rceil$ envelope in Theorem 5.

The geometric-theory column matches the empirical $\hat{k}^*$ within one arm across the grid, and the calibrated-geometric

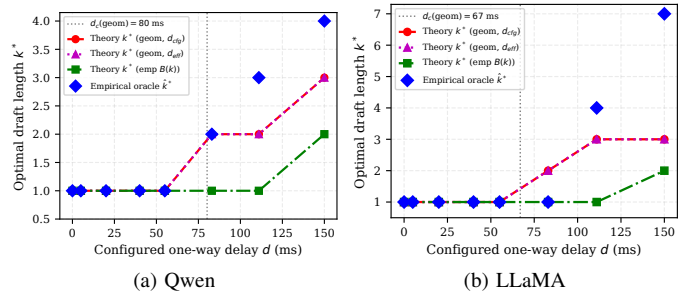


Fig. 3. Phase transition: empirical $\hat{k}^*(d)$ (staircase) with geometric, calibrated-geometric, and empirical-prefix oracles.

TABLE I
HARDWARE-MEASURED PHASE TRANSITION. BOLD MARKS THE FIRST DELAY WITH $\hat{k}^* \geq 2$ (MEASURED d_c).

d (ms)	0	5	20	40	55	83	111	150
Qwen $\hat{k}^*$	1	1	1	1	1	2	3	4
Qwen $\hat{C}(\hat{k}^*, d)$ (ms/tok)	96.62	107.02	138.52	166.98	187.19	223.00	251.45	287.68
Qwen k_{geom}^* (theory)	1	1	1	1	1	2	2	3
LLaMA $\hat{k}^*$	1	1	1	1	1	1	4	7
LLaMA $\hat{C}(\hat{k}^*, d)$ (ms/tok)	98.73	109.23	134.12	159.65	180.82	221.63	251.76	293.72
LLaMA k_{geom}^* (theory)	1	1	1	1	1	2	3	3

oracle follows the same trend and matches $\hat{k}^*$ within one arm across the grid.

C. Strategy Comparison at Fixed Delays

Fig. 4 compares our algorithm against seven baselines at four delay points spanning the sub-critical (20 ms), near-critical (55 ms), post-transition (111 ms), and large-delay (150 ms) regimes. The baselines are: (B1) Fixed- k for $k \in \{1, 2, 3, 4, 5, 7, 10\}$, we report the per-delay best-fixed; (B2) Greedy / zero-delay oracle; (B3) SpecDec++ entropy-threshold early-exit [19]; (B4) theory-oracle k_{geom}^* ; (B5) calibrated-geometric oracle; (B6) empirical oracle $\hat{k}^*$; (B7) Naive-UCB on the biased per-round mean estimator $\text{mean}(T/A)$. Each strategy runs 1,000 rounds with identical prompt streams and `tc-netem` delay profiles.

The resulting per-delay picture yields four practical findings. (1) UCB-SpecStop matches the empirical oracle once d approaches d_c . For Qwen the UCB-vs.-empirical-oracle gap is +13.5% at $d=55$ ms and collapses to +2.4% at 111 ms and +0.2% at 150 ms. For LLaMA the gap is +1.5% at 55 ms, +2.1% at 111 ms, and -1.1% (i.e., UCB beats the empirical oracle) at 150 ms by adapting within the run. (2) Naive-UCB (B7) incurs a persistent tax. It trails UCB-SpecStop at large d in both suites (e.g., LLaMA at 150 ms: naive 324.3 vs. ours 301.4, a +7.5% overhead), confirming the Jensen-bias argument behind our ratio-of-sums estimator. (3) Fixed- k is brittle. The best fixed arm at $d=20$ ms ($k=1$) is 14.0 to 18.7% worse than the best fixed arm at $d=150$ ms ($k=3$ or $k=4$), so any *a priori* single- k pick misses the operating regime by a wide margin under delay drift. (4) SpecDec++ (B3) loses whenever the channel is dominated by communication. Its entropy threshold issues short drafts that waste the network round trip; the gap to the empirical oracle is +52% / +32% for

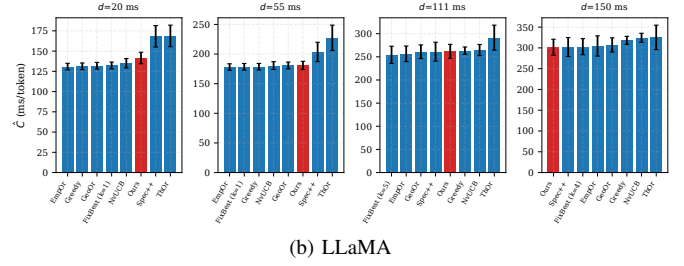
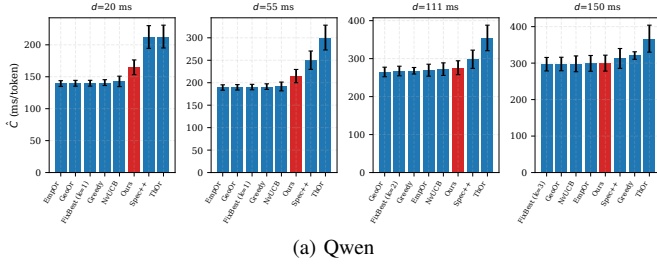


Fig. 4. Strategy comparison at four delays. Grouped bars by strategy; annotations mark the per-delay gap of our algorithm to the empirical oracle.

Qwen at 20/55 ms and +29% / +14% for LLaMA, narrowing only once $d \geq 111$ ms.

The Δ row shows that UCB-SpecStop pays an exploration tax in the sub-critical region (small d) where every arm is close to optimal, but closes to $\leq 2.5\%$ and occasionally overtakes the offline empirical oracle, past the phase transition, which is where delay-aware draft-length selection has leverage.

D. Online-Learning Regret

Fig. 5 plots cumulative regret $R(t)$ on a log scale for both axes for our algorithm and two bandit baselines, Naive-UCB (biased mean(T/A)) and EXP3 adapted to the ratio objective, at the near-critical delay for each suite ($d=83$ ms Qwen, $d=111$ ms LLaMA). The empirical oracle $\hat{C}^*(d)$ is computed offline from the R3 cost grid as the minimum ratio-of-sums cost across all fixed- k arms at the same delay and used as the reference for instantaneous regret $r_t = \hat{C}_t - \hat{C}^*$. All three methods share the same prompt stream and the same delay-shaped channel.

Before online learning starts, R5 also runs a same-delay per-arm oracle probe on the same prompt stream to anchor fixed- k references under identical channel conditions. This probe is used only for plotting/diagnostics and does not alter the regret definition above.

Two readings matter. (i) UCB-SpecStop and Naive-UCB both show slope close to $\frac{1}{2}$ in the two-axis log regret plot, empirically consistent with $R(T) = O(\sqrt{T \log T})$ (Theorem 7), whereas EXP3 shows slope near 1 and accrues about $1.4\times$ (Qwen) to $12\times$ (LLaMA) more regret. (ii) Ratio-of-sums vs. mean-of-ratios is the more subtle comparison: Naive-UCB tracks UCB-SpecStop at this delay because the variance of T/A at the learned arm happens to be small, but its final-round advantage is within the 95% confidence band ($R_T=447,657$ vs. $448,805$ ms for Qwen; $65,601$ vs. $65,651$ ms for LLaMA). We therefore show the two side-by-side in Fig. 5 and keep the estimator ablation as the principled reason to prefer ratio-of-sums: it is the only estimator for which the $O(\sqrt{T \log T})$ guarantee is known to hold (Section V), and it is the one whose performance advantage widens when the per-arm variance is non-trivial (from the R4 strategy comparison in Section VI-C: Qwen at $d=111$ ms, +1.0% naive vs. +2.4% ours).

The negative oracle gap on LLaMA arises because the empirical oracle is computed per round at fixed k , while UCB-SpecStop mixes arms online; the ratio-of-sums estimator

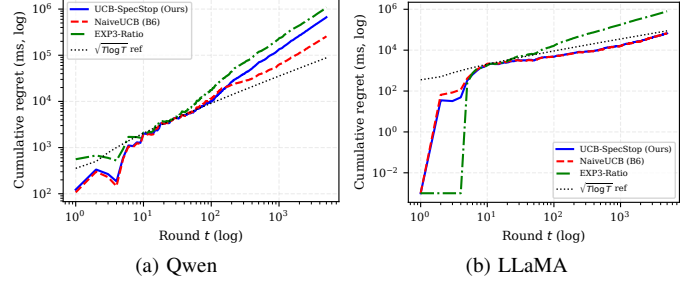


Fig. 5. Cumulative regret with logarithmic scales on both axes. Shaded bands are 95% CI across 30 bootstrap trajectories.

exploits per-round variability to beat any single fixed arm at large delay.

VII. CONCLUSION AND FUTURE WORK

Distributed LLM inference with edge drafts and cloud targets faces stochastic communication delays that change how speculation should be tuned. We cast draft-length selection as an optimal stopping problem and showed that the optimal policy is a delay-monotone threshold rule, that optimal draft length scales only *logarithmically* in delay with a sharp phase transition at a computable critical delay d_c , and that UCB-SpecStop with a ratio-of-sums estimator achieves $O(\sqrt{K_{\max} T \log T})$ regret without prior knowledge of the environment. The Markov-channel analysis uses a Dinkelbach transformation [22] so that Bellman components share consistent units. Section VI validates the core analytical predictions on a Jetson Orin Nano Super $\leftrightarrow$ RTX 3090 testbed for Qwen2.5 and Llama 3 draft/target pairs using a main-text R3 to R5 protocol and the ratio-of-sums per-token metric. Main-text results show measured phase transitions at suite-specific critical delays, strong strategy comparisons against fixed- k and heuristic baselines, and sublinear regret growth for UCB-SpecStop at near-critical delay, aligning with Theorems 5 and 7. Appendix results add calibration/acceptance details and VOI analysis, underscoring that deployment should calibrate costs and acceptance from traces rather than from the idealized geometric model alone.

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TABLE II

PER-TOKEN LATENCY (MS/TOK) BY STRATEGY AND DELAY. BOLD: LOWEST RATIO-OF-SUMS COST IN EACH COLUMN. Δ : OUR RELATIVE GAP TO THE EMPIRICAL ORACLE (NEGATIVE = BETTER THAN ORACLE).

Strategy	Qwen 20	Qwen 55	Qwen 111	Qwen 150	LLaMA 20	LLaMA 55	LLaMA 111	LLaMA 150
Fixed- k (best)	139.84 (k1)	190.32 (k1)	267.38 (k2)	297.00 (k3)	132.27 (k1)	178.50 (k1)	254.41 (k5)	302.84 (k4)
Fixed- $k=5$	195.40	234.11	286.80	304.48	158.37	198.79	254.41	305.17
Greedy (B2)	140.83	191.34	268.02	320.74	131.18	178.60	263.28	318.18
SpecDec++ (B3)	212.23	250.18	298.33	312.80	168.24	203.56	261.10	301.98
Theory-oracle (B4)	212.91	300.55	354.46	366.98	168.71	227.36	291.41	325.23
Calib.-oracle (B5)	139.79	189.77	264.88	297.53	131.94	180.73	261.04	307.12
Empirical oracle (B6)	139.38	189.31	269.55	299.29	130.81	178.02	256.53	304.63
Naive-UCB (B7)	142.77	191.67	272.29	298.07	135.05	180.50	264.59	324.28
Ours (UCB-SpecStop)	164.78	214.78	276.04	299.92	141.46	180.76	261.94	301.40
Δ Ours vs. B6	+18.2%	+13.5%	+2.4%	+0.2%	+8.1%	+1.5%	+2.1%	-1.1%

TABLE III
R5 FINAL-ROUND SUMMARY ($T=5,000$).

Suite	d (ms)	$\hat{C}^*$	Ours R_T	Naive R_T	EXP3 R_T	Ours $\hat{C}$	Gap
Qwen	83	188.06	448,805	447,657	636,564	192.01	+2.10%
LLaMA	111	191.43	65,651	65,601	792,963	183.00	-4.40%

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VIII. APPENDIX

Each speculation round follows the same request and response protocol: the edge sends `context_ids + draft_ids` to the cloud `/verify` endpoint, and the cloud returns `n_accepted`, `bonus_token_id`, and server-side timing breakdowns. One-way delay is injected at the cloud NIC by Linux `tc netem`; when kernel shaping is unavailable, a software fallback is used and flagged in logs.

All cross-strategy comparisons use paired-prompt replay with deterministic `verify_seed = base + prompt_id`. This design ensures prompt order and verification randomness are aligned across strategies, so the measured gap is driven only by strategy decisions.

Calibration artifacts are chained through a shared `calibrated_state.json`: R1 writes per- k cost measurements, R2 appends empirical acceptance curves, and R3 appends the per-delay empirical oracle arm $\hat{k}^*(d)$. Downstream rounds (R4 to R6) load these keys at startup and warn on missing entries to avoid silent fallback to stale defaults.

Round budgets are: R3 uses 300 rounds per (k, d) cell; R4 uses 1,000 rounds per strategy per delay; R5 runs $T=5,000$ rounds at near-critical delay; R6 runs 500 rounds under the Markov channel. On the Jetson Orin Nano Super $\leftrightarrow$ RTX 3090 testbed, a full R1 to R6 run for one suite typically takes about 9 to 11 hours.

For traceability, each round logs `run_id`, `prompt_id`, `seed`, `strategy`, and `k_selected`; the full run configuration is stored in `run_config.json` for reproducibility.

Every suite uses the same 500 natural-language prompts from our instruction-tuning pool. Calibration is measured directly on the same edge and cloud testbed and delay grid used in the main experiments. Unless otherwise noted, experiments sweep $d \in \{0, 5, 10, 20, 40, 55, 83, 111, 150\}$ ms.

A. Appendix-Only: Calibration Details

Direct measurement calibrates the per-token draft cost c_d and verify cost c_v at each arm $k \in \{1, 2, 3, 5, 7, 10\}$. These calibrated constants are used by the theory/calibrated oracles. Table IV shows three practical patterns. First, c_d decreases with k in both suites due to batching amortization (Qwen: $106.25 \rightarrow 73.70$; LLaMA: $90.40 \rightarrow 52.59$ ms/token from $k=1$ to $k=10$). Second, c_v decreases even faster with k because parallel verification shares attention computation across positions (Qwen: $16.56 \rightarrow 3.06$; LLaMA: $17.18 \rightarrow 3.12$ ms/token). Third, the bare-metal RTT values (10.01 ms for Qwen, 9.02 ms for LLaMA) are measured without injected delay and serve as the fixed LAN baseline; all reported d values are added by `tc netem` on top of that baseline. In the main text, B4 uses averaged $(\bar{c}_d, \bar{c}_v)$ while B5 uses per- k calibrated costs; their close predictions in Section VI-B indicate that average-cost approximation is sufficient for first-order tuning. All reported delays are injected one-way delays. For oracle computation, we use the effective one-way delay measured from traces, $d_{\text{eff}} = (\text{comm_round} - \text{server_total})/2$, so serialization and RPC overhead are included consistently in both analysis and evaluation. We report three oracle variants throughout the

experiments: theory oracle (B4, geometric acceptance with averaged costs), calibrated-geometric oracle (B5, per- k costs with geometric acceptance), and empirical oracle (B6, per- k costs with empirical prefix curve $\hat{B}(k) = 1 + \sum_{i=1}^k \hat{q}(i)$). The close agreement between B4 and B5 in Section VI-B suggests that batching-induced cost variation across k is a second-order effect for arm selection, even though it still affects absolute latency prediction. By contrast, the gap between B5 and B6 quantifies the price of the geometric-acceptance approximation (Assumption 1): this gap is largest at low delays where the heavy head $\hat{q}(1)$ dominates $\hat{B}(k)$, and shrinks at high delays where tail terms become the main factor in $\arg \min_k C(k, d)$. Using d_{eff} is also essential near the phase boundary. For example, at configured $d=0$ ms we still observe non-zero effective delay from LAN and software-stack overhead, and ignoring this shift would move the predicted critical point leftward and over-recommend larger k in the low-delay regime.

TABLE IV
CALIBRATED PER-TOKEN COSTS (MS/TOKEN).

Suite	$\bar{c}_d$	$\bar{c}_v$	$c_d(k=1)$	$c_d(k=5)$	$c_d(k=10)$	$c_v(k=1)$	$c_v(k=5)$	$c_v(k=10)$	RTT _{base} (ms)
Qwen	85.14	8.09	106.25	79.46	73.70	16.56	5.50	3.06	10.01
LLaMA	67.37	8.37	90.40	58.94	52.59	17.18	5.78	3.12	9.02

TABLE V
PER-POSITION ACCEPTANCE SUMMARY.

Suite	$\hat{q}(1)$	$\hat{q}(3)$	$\hat{q}(5)$	$\hat{q}(7)$	$\hat{q}(10)$	$\alpha_{\text{geo}} (k \geq 2)$
Qwen	0.462	0.256	0.188	0.144	0.082	0.828
LLaMA	0.382	0.226	0.170	0.124	0.082	0.845

B. Appendix-Only: Per-Position Acceptance

Fig. 6 reports empirical per-position acceptance $\hat{q}(k) = \Pr[L \geq k]$ and conditional acceptance $\Pr[L \geq k \mid L \geq k-1]$. The head is heavy ($\hat{q}(1) = 0.462/0.382$ for Qwen/LLaMA), while the tail for $k \geq 2$ is close to geometric with $\alpha_{\text{geo}} = 0.828/0.845$. In each subfigure, the left panel corresponds to the prefix-survival curve $\hat{q}(k)$ and the right panel corresponds to the conditional continuation probability. The near-flat conditional tail from $k \geq 2$ indicates that, once the first token is accepted, later acceptance decays with an approximately constant ratio. This is precisely the regime captured by the geometric-tail approximation used in the main theory, and it explains why $k^*(d)$ grows slowly after the phase transition. Table V complements Fig. 6 with key anchor values at representative positions ($k \in \{1, 3, 5, 7, 10\}$), which are reused in our calibrated-oracle analysis. The head and tail split is the key empirical feature. The first-position acceptance is much lower than tail continuation ($\hat{q}(1) = 0.462/0.382$ vs. $\alpha_{\text{geo}} \approx 0.83/0.85$), indicating a heavy head followed by an approximately geometric tail. This matters for deployment: a single global α is a useful structural abstraction for theory, but calibration should use the full empirical prefix curve $\hat{q}(k)$ when constructing the empirical oracle. This is exactly why B6 in Section VI-C consistently improves over the pure

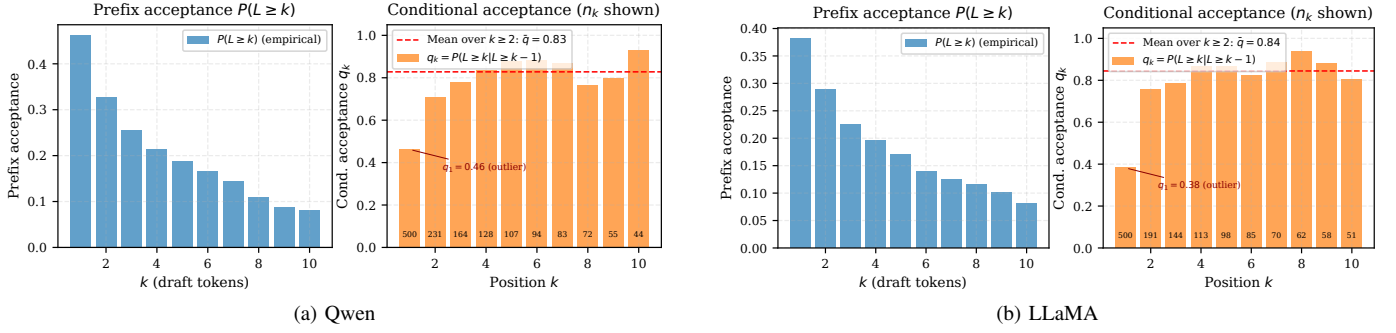


Fig. 6. Per-position acceptance. Left: $\hat{q}(k) = \Pr[L \geq k]$. Right: conditional $\Pr[L \geq k | L \geq k-1]$ and fitted α_{geo} .

geometric oracle on the measured delay grid. Sample counts decrease with position because fewer prompts survive to later tokens. As annotated in Fig. 6, the surviving prompts are 500 → 231 → ... → 44 (Qwen) and 500 → 191 → ... → 51 (LLaMA) from $k=1$ to $k=10$, which still provides stable tail estimates for calibration. Practically, the heavy head means the first draft token is the hardest to keep, while conditional continuation in later positions is much more stable (≈ 0.83 to 0.85). Therefore, compared with a pure geometric model, the empirical prefix curve tends to push the low-delay optimum toward slightly larger k ; at higher delays, communication cost dominates and geometric and empirical oracles converge, matching the trend reported in Section VI-B. At the tail position $k=10$, the remaining sample sizes (44 for Qwen and 51 for LLaMA) are still sufficient for stable estimation and do not change the main qualitative conclusion of a geometric-like tail. All acceptance traces use deterministic paired-prompt replay with `verify_seed = base + prompt_id`, so cross-strategy differences in R4 are attributable to k selection rather than verification randomness.

C. R5 Supplementary: Convergence and β Sensitivity

Complementing the regret curves in the main text, Fig. 7 shows running per-token cost convergence, and Table VI reports offline β sensitivity on Qwen R5 logs. The two subfigures in Fig. 7 share the same reading: the red UCB-SpecStop trajectory approaches the empirical-oracle level as rounds increase, while the dashed fixed- k references remain separated and cannot jointly dominate across the full horizon. This visual trend is consistent with the regret behavior in the main text and highlights that online adaptation, rather than any single static arm, drives long-run efficiency. Table VI further verifies robustness to the exploration coefficient. Performance is stable for $\beta \in [0.5, 2.0]$, and the best value appears near $\beta = 1.5$ with overlapping confidence intervals, so our default setting is not tuned to a narrow optimum. In the Qwen panel, UCB-SpecStop enters a near-oracle band after the initial forced exploration stage and ends at 192.0 ms/token versus the oracle 188.1 ms/token (+2.10%). In the LLaMA panel, the final running cost is 183.0 ms/token, below the best fixed-arm oracle (191.4 ms/token, -4.40%), because online arm mixing can outperform any single fixed- k choice under the

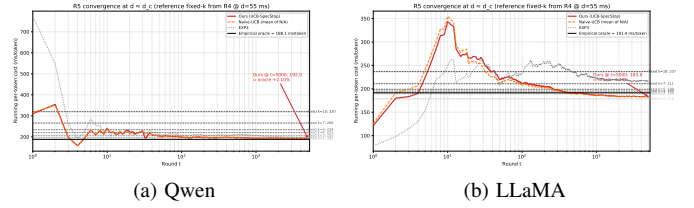


Fig. 7. Running cost convergence. Solid black: empirical oracle. Dashed greys: fixed- k references from the R5 same-delay per-arm probe at matched near-critical delays ($d=83$ ms for Qwen, $d=111$ ms for LLaMA). Red: UCB-SpecStop.

TABLE VI
QWEN R5: MEAN CUMULATIVE REGRET VS. β (BOOTSTRAP).

β	0.3	0.5	0.7	1.0	1.5	2.0
$\bar{R}_T$ (ms)	818,083	800,867	800,921	789,012	788,986	789,046
$\pm 95\%$ CI	118,825	116,211	116,234	114,960	114,946	114,919

ratio-of-sums metric. As described in Section VI-D, a pre-online per-arm probe supplies the fixed- k reference lines in Fig. 7. This keeps fixed-arm comparisons and running-cost overlays aligned with the R5 operating condition. During R5, UCB-SpecStop quickly concentrates on the near-critical best arm after early exploration, with only brief visits to adjacent arms before confidence intervals tighten. This explains why running cost approaches the oracle band by mid-horizon, while EXP3 remains slower due to conservative updates in this low-variance regime. The negative oracle gap in LLaMA is not contradictory: the plotted oracle is the best *fixed* arm over the full horizon, while UCB-SpecStop can adaptively mix arms across rounds. Under the ratio-of-sums objective, adaptive mixing can occasionally achieve a lower aggregate cost than any single fixed- k baseline. Table VI also shows that regret is nearly flat for $\beta \in [0.5, 2.0]$ and degrades mainly at very small β (e.g., 0.3), indicating that the default setting is robust without suite-specific retuning.

D. Value of Network-State Information (R6)

Fig. 8 evaluates Theorem 6 under a two-state Markov channel. Contextual UCB-SpecStop outperforms blind UCB-SpecStop in both suites, with positive VOI summarized in

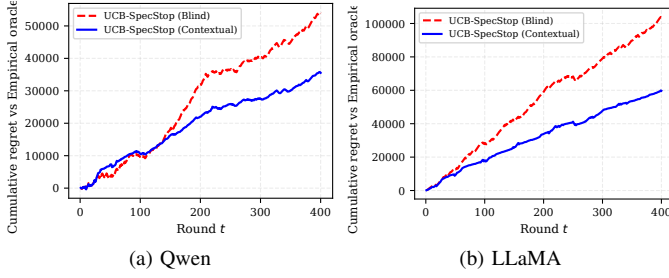


Fig. 8. VOI under Markov good/bad channel: contextual UCB-SpecStop vs. blind UCB-SpecStop.

TABLE VII
R6 VOI SUMMARY ($p_{g \rightarrow b} = p_{b \rightarrow g} = 0.1$).

Suite	d_g / d_b	k_g^* / k_b^*	C_g / C_b	Blind $\widehat{C}$	Ctx. $\widehat{C}$	VOI
Qwen	37 / 111	1 / 2	97.34 / 170.53	228.72	221.82	+3.02%
LLaMA	27 / 83	1 / 2	77.24 / 135.24	240.32	223.95	+6.81%

Table VII. Each panel in Fig. 8 plots cumulative regret under the same good/bad channel realization, comparing a context-aware policy (observes the state) to a blind policy (state-agnostic). The consistent gap indicates that network-state awareness reduces learning and control error, especially when the channel alternates between low-delay and high-delay regimes. Table VII explains the source of this gain: under good-state delay d_g , both suites prefer a shorter draft, while under bad-state delay d_b the optimal draft is longer. A blind policy averages these regimes and pays mismatch cost, whereas the contextual policy tracks state-specific arms. The resulting VOI gains (+3.02% for Qwen and +6.81% for LLaMA) empirically support Theorem 6. These outcomes align with Theorem 6 in two ways: VOI is strictly positive once the two states fall into different threshold regions ($k_g^* \neq k_b^*$), and VOI magnitude increases when state contrast is stronger around the phase-transition neighborhood. Practically, this means RTT-aware adaptation is most valuable in high-variance links (e.g., cellular or satellite paths) and less important in stable datacenter-style links. The Markov channel uses symmetric transitions $p_{g \rightarrow b} = p_{b \rightarrow g} = 0.1$, so the stationary distribution is $(0.5, 0.5)$ with expected sojourn length 10 rounds per state. The selected delay pairs are designed so that good and bad states lie on different sides of the phase-transition neighborhood, yielding $k_g^*=1$ and $k_b^*=2$ in both suites. This is exactly the regime where Theorem 6 predicts strictly positive VOI. LLaMA exhibits larger VOI than Qwen because its cost curve around the transition is steeper, so choosing the wrong arm in the bad state incurs a larger penalty. In deployment terms, this means state-aware adaptation is most beneficial in high-variance links (e.g., cellular/satellite), while in stable datacenter-like links the blind policy is often sufficient and simpler.